Using iNaturalist, eBird, and Python to become a community scientist

Richard Littauer



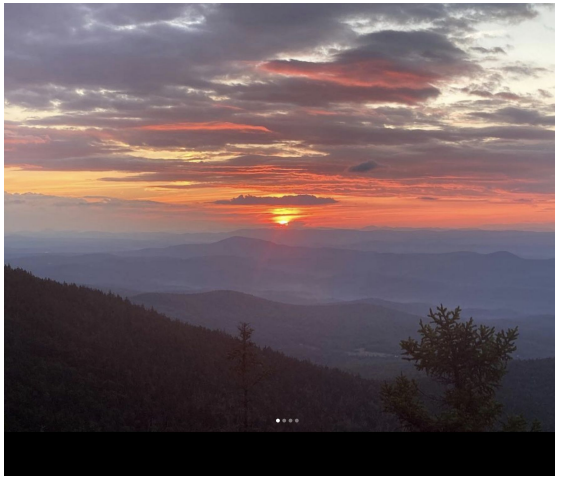
Matuku moana

Content warning

Bugs, dinosaurs, genocide, privilege

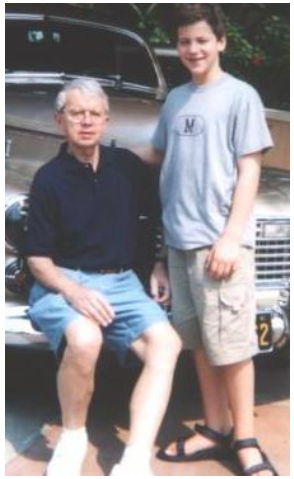
Ko wai a Richard Littauer, ahakoa?

- Ko Richard tōku inga
- No Europe oku tupuna
- Kei te noho au ki Te Whanganui-a-Tara.











Who is this Richard Littauer, anyway

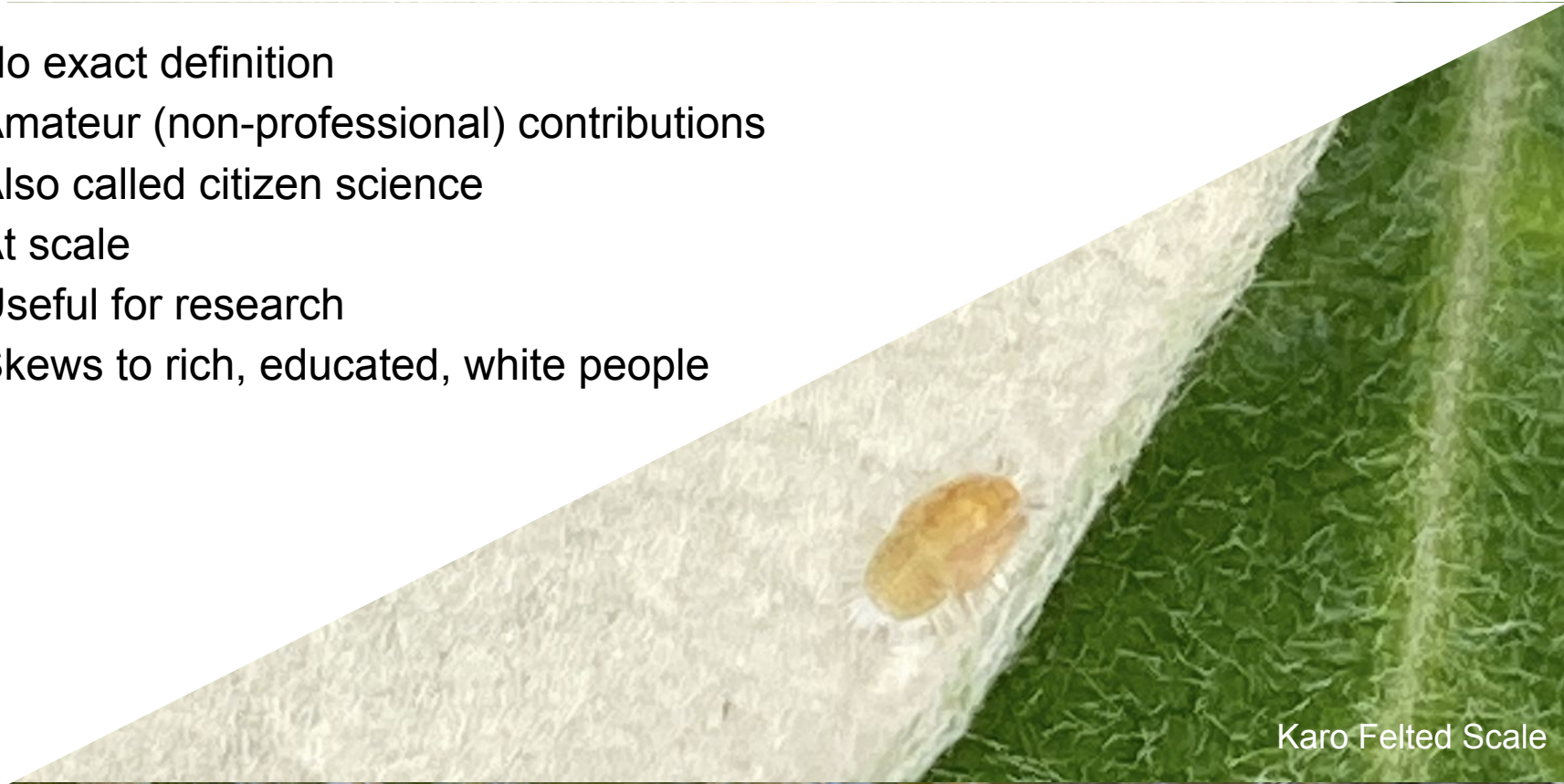
- "Normal person"
- Linguist
- Software Developer
- Open sourcerer
- Currently:
  - **SustainOSS**
  - **CURIOSS**
  - **GNOME**
  - **Vic**





Community Science

- No exact definition
- Amateur (non-professional) contributions
- Also called citizen science
- At scale
- Useful for research
- Skews to rich, educated, white people



SCIENCE

- Silver Fern, collected by Banks and Solander
- C. 1769-1770, Aotearoa
- Specimen in Te Papa



SCIENCE

- Silver Fern, photographed by Littauer
- Nov 14, 2022, Ohakune
- https://www.inaturalist.org/observations/141999533



SCIENCE

eBird New Zealand CHECKLIST S122400012

Mon **14 Nov 2022** 7:06 PM

Edit date and effort

Tongariro NP -- Maungawhero Forest Walk

Ruapehu District County, Manawatu-Wanganui, New Zealand

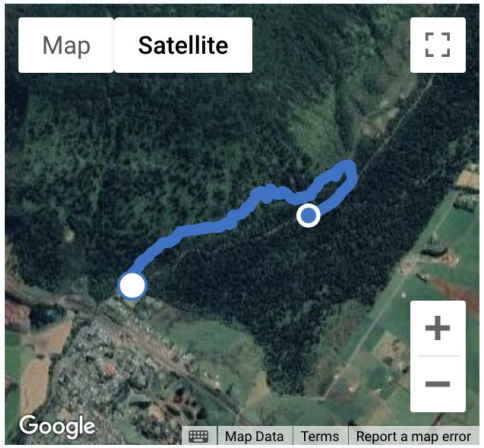
Edit location

Richard Littauer

Share

Traveling Incomplete

2 1 hr, 3 min 1.33 mi



eBird Mobile Tracks

Submitted from eBird for iOS, version 2.13.20

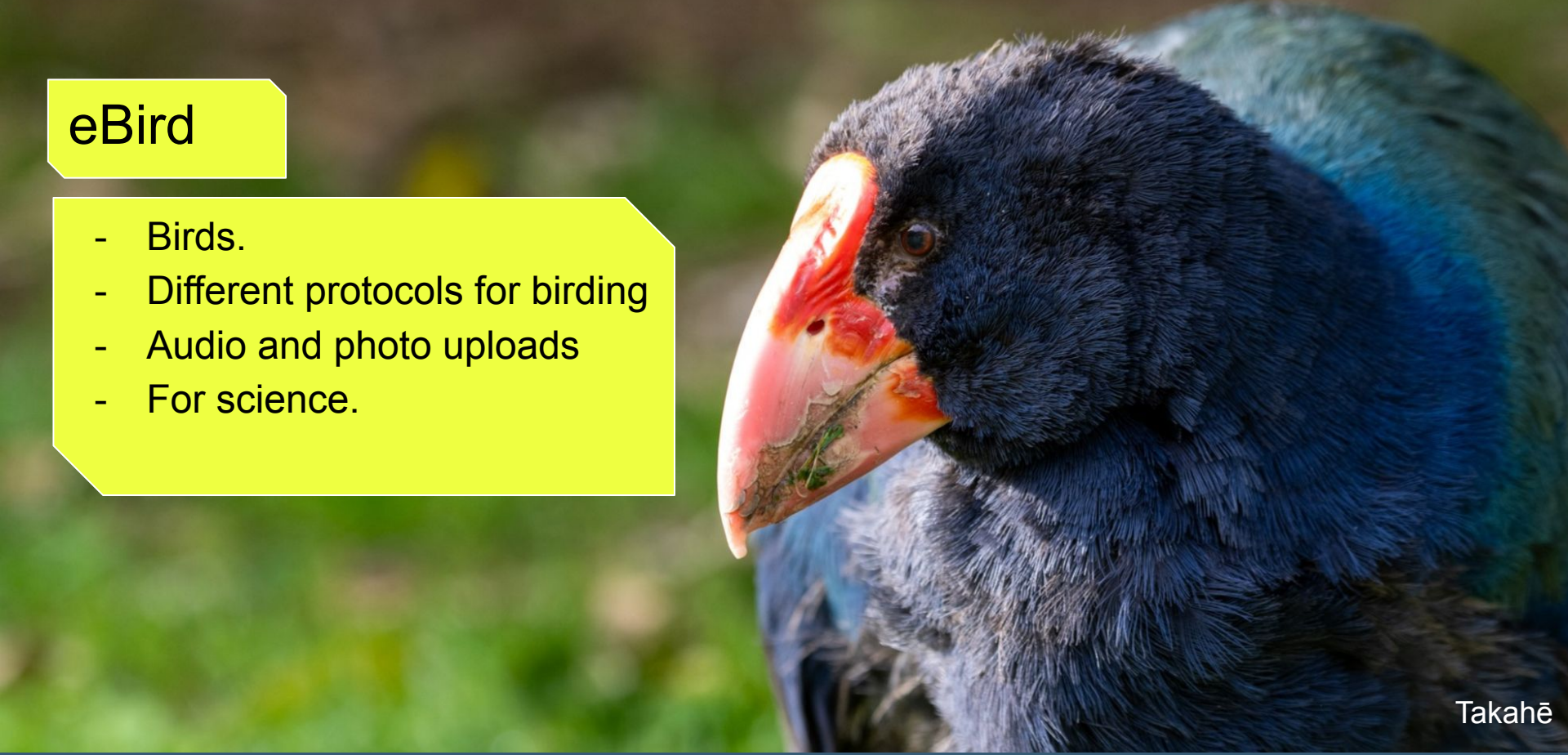
Edit comments

8 Species Observed 16 individuals

1	**New Zealand Pigeon**	*Hemiphaga novaeseelandiae*
1	**Long-tailed Koel**	*Urodynamis taitensis*
1	**Tui**	*Prosthemadera novaeseelandiae*
2	**Gray Gerygone**	*Gerygone igata*
5	**Tomtit**	*Petroica macrocephala*
3	**Silvereye**	*Zosterops lateralis*
1	**Song Thrush**	*Turdus philomelos* *
2	**Eurasian Blackbird**	*Turdus merula* *

eBird

- Birds.
- Different protocols for birding
- Audio and photo uploads
- For science.



eBird Stats WORLD

10,826
Species

97,753,170
Complete Checklists

1,034,384
eBirders

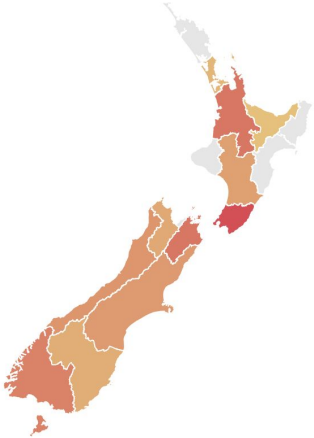
eBird

- Regions
- Hotspots



eBird

- Users
- Alerts



Cirl Bunting

eBird

- Reviewers



iNaturalist

- All momo
- Photos or audio
- "Research grade"
- Comments



Korimako, Manuka, & Rō

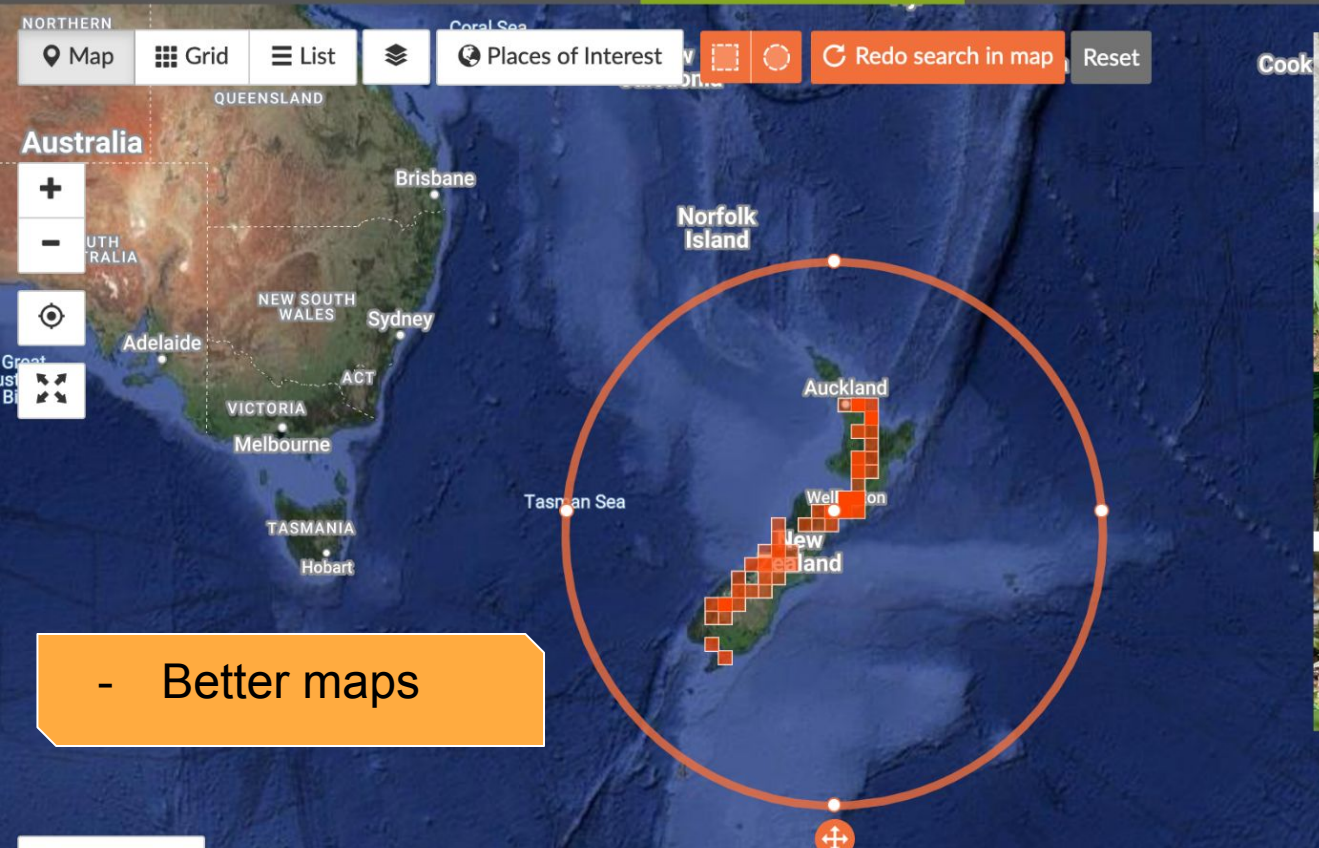
Observations

Species | Location | Go | Filters

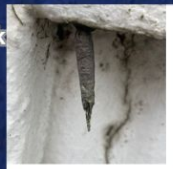
Custom Boundary

3,286 OBSERVATIONS | 875 SPECIES | 340 IDENTIFIERS | 1 OBSERVER

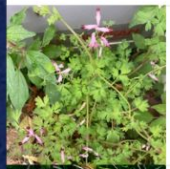
Map | Grid | List | Places of Interest | Redo search in map | Reset



- Better maps



New Zealand Common Bag Moth
(*Liothula omnivora*)
North Island / Te... • Today
Research Grade 2 8h



Common Ramping-Fumitory
(*Fumaria muralis*)
North Island / Te... • Today
Research Grade 2 8h



New Zealand Sacred Kingfisher
(*Todiramphus sanctus vagans*)
North Island / Te... • Today
Research Grade 3 8h



Garden Snail
(*Cornu aspersum*)
North Island / Te... • Yesterday
Research Grade 2 1d

Observations



Pythons

New Zealand

Go

Filters 3

New Zealand

0 OBSERVATIONS | 0 SPECIES | 0 IDENTIFIERS | 0 OBSERVERS

Map | Grid | List | Places of Interest | Redo search in map

No results found

Map Legend

Keyboard shortcuts | Map data



Thank you.

Richard Littauer – burntfen.com

Finding eBird User Checklists

- Sometimes useful for seeing what others are doing



Toroa, Pekehā

Finding eBird User Checklists

```python
# Get all of the checklist IDs for a user from an eBird download.
# Useful for finding all checklists from a specific person.
#
# USAGE
# Copy this file into an ebd_ folder, and then run $ py find... obsrID
######################

import pandas as pd
import glob
import sys

# Get observer ID from the command line arguments
if len(sys.argv) != 2:
    print("Usage: script.py <observer_id>")
    sys.exit(1)

observer_id = sys.argv[1]

# Find all files matching the pattern ebd_*.txt
file_pattern = "test.txt"
files = glob.glob(file_pattern)

# Process each file
for file_path in files:
    print(f"Processing file: {file_path}")

    # Load the file
    df = pd.read_csv(file_path, delimiter='\t')

    # Filter rows where "OBSERVER ID" matches the provided observer ID
    filtered_df = df[df["OBSERVER ID"] == observer_id]

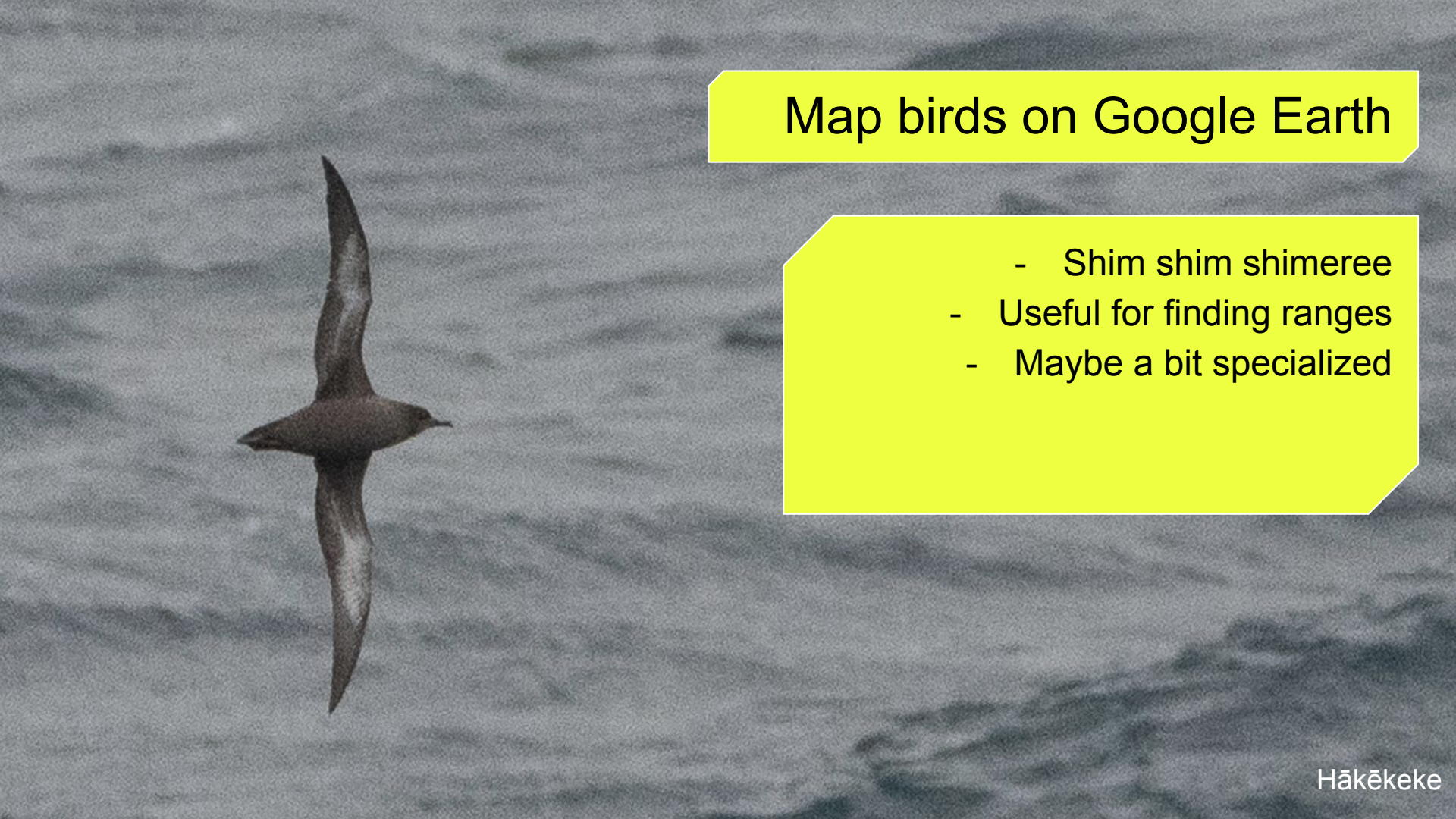
    # Get unique entries in the "SAMPLING EVENT IDENTIFIER" column
    unique_common_names = filtered_df["SAMPLING EVENT IDENTIFIER"].unique()

    # Print the unique entries
    for name in unique_common_names:
        print("https://ebird.org/checklist/" + name)
```

- Super simple
- Needs eBird export
- Python because good

Pekehā

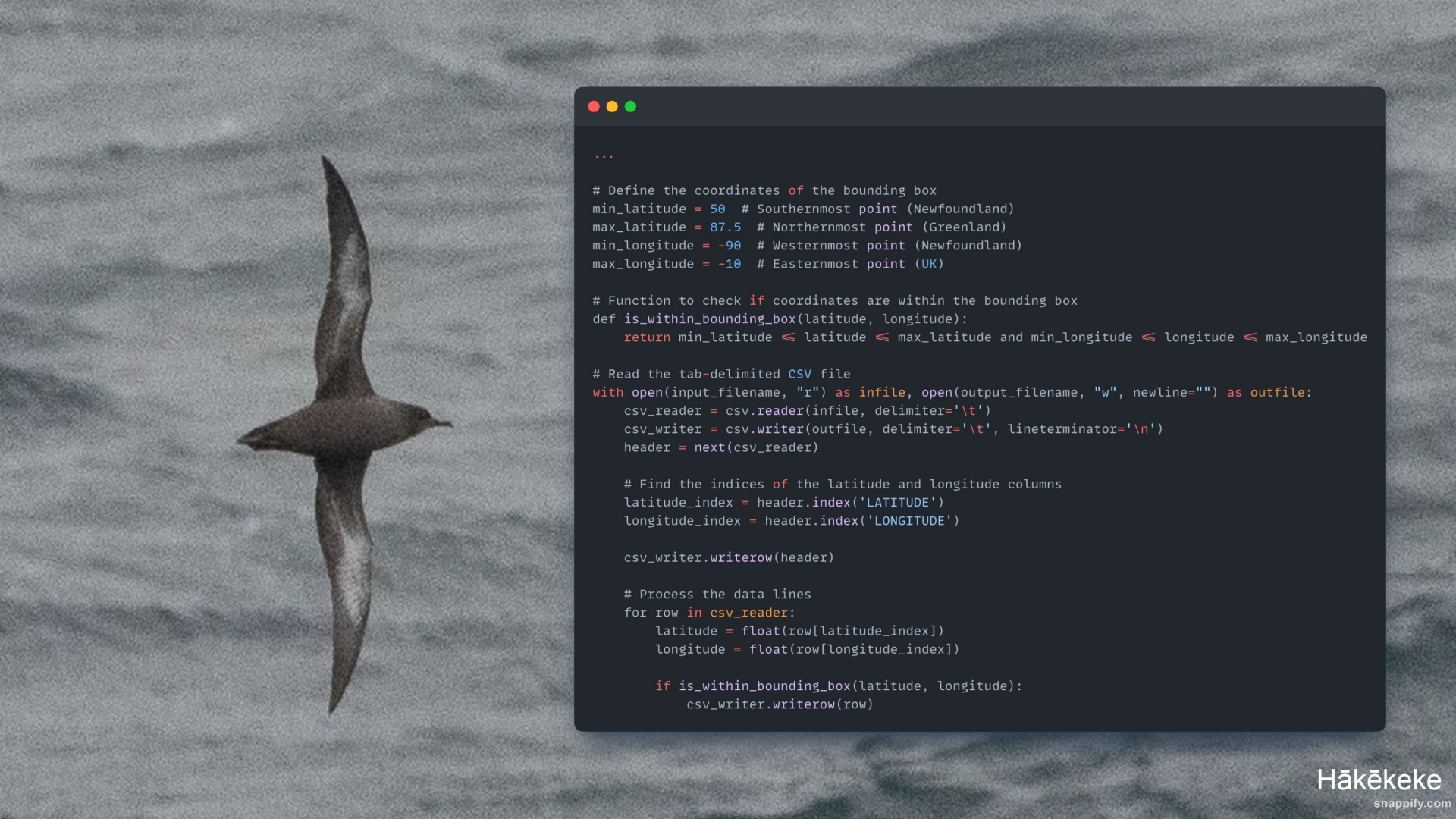
snappify.com



Map birds on Google Earth

- Shim shim shimeree
- Useful for finding ranges
- Maybe a bit specialized

Hākēkeke



```
...

# Define the coordinates of the bounding box
min_latitude = 50  # Southernmost point (Newfoundland)
max_latitude = 87.5  # Northernmost point (Greenland)
min_longitude = -90  # Westernmost point (Newfoundland)
max_longitude = -10  # Easternmost point (UK)

# Function to check if coordinates are within the bounding box
def is_within_bounding_box(latitude, longitude):
    return min_latitude ⩽ latitude ⩽ max_latitude and min_longitude ⩽ longitude ⩽ max_longitude

# Read the tab-delimited CSV file
with open(input_filename, "r") as infile, open(output_filename, "w", newline="") as outfile:
    csv_reader = csv.reader(infile, delimiter='\t')
    csv_writer = csv.writer(outfile, delimiter='\t', lineterminator='\n')
    header = next(csv_reader)

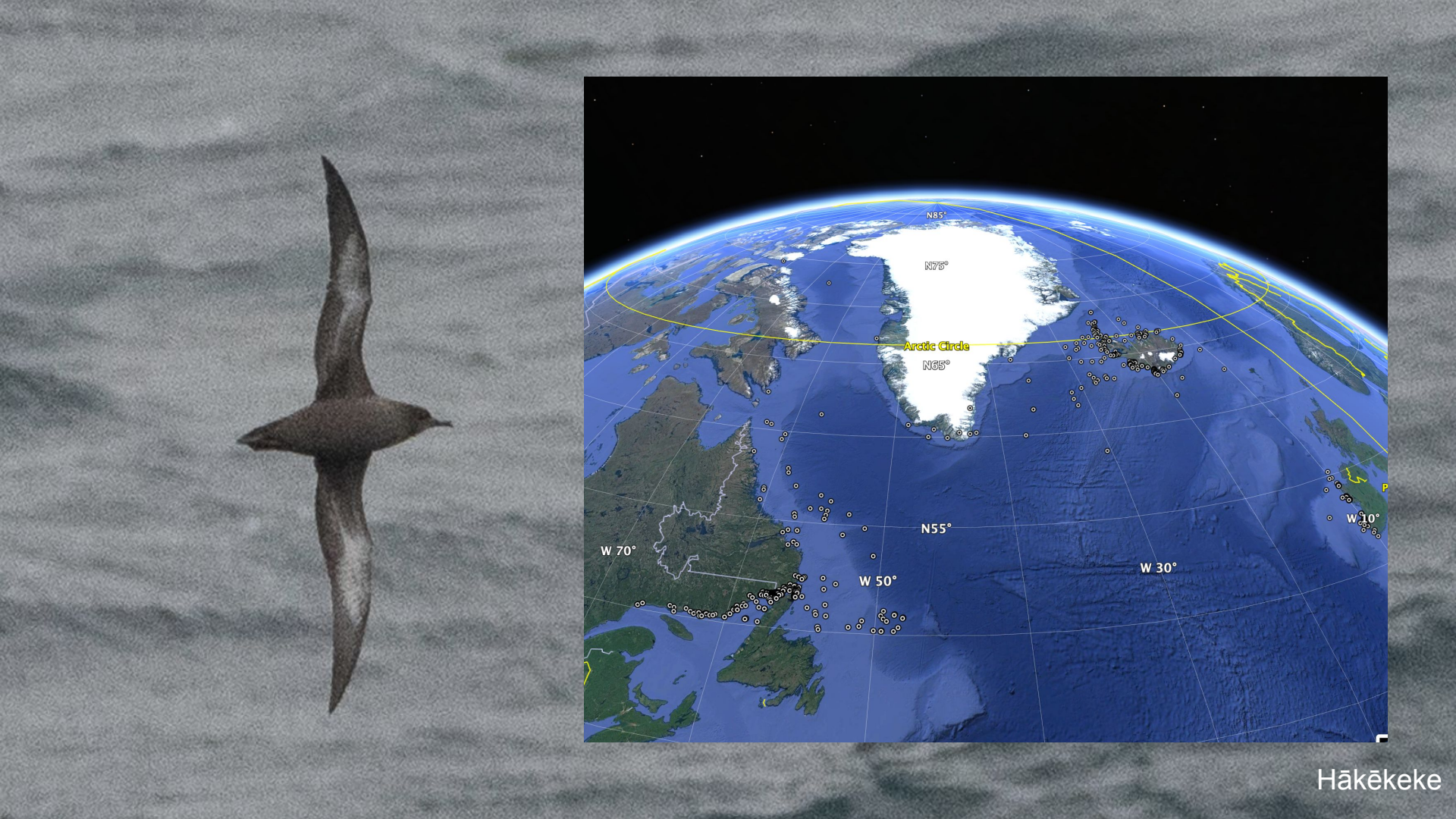
    # Find the indices of the latitude and longitude columns
    latitude_index = header.index('LATITUDE')
    longitude_index = header.index('LONGITUDE')

    csv_writer.writerow(header)

    # Process the data lines
    for row in csv_reader:
        latitude = float(row[latitude_index])
        longitude = float(row[longitude_index])

        if is_within_bounding_box(latitude, longitude):
            csv_writer.writerow(row)
```

Hākēkeke
snappify.com



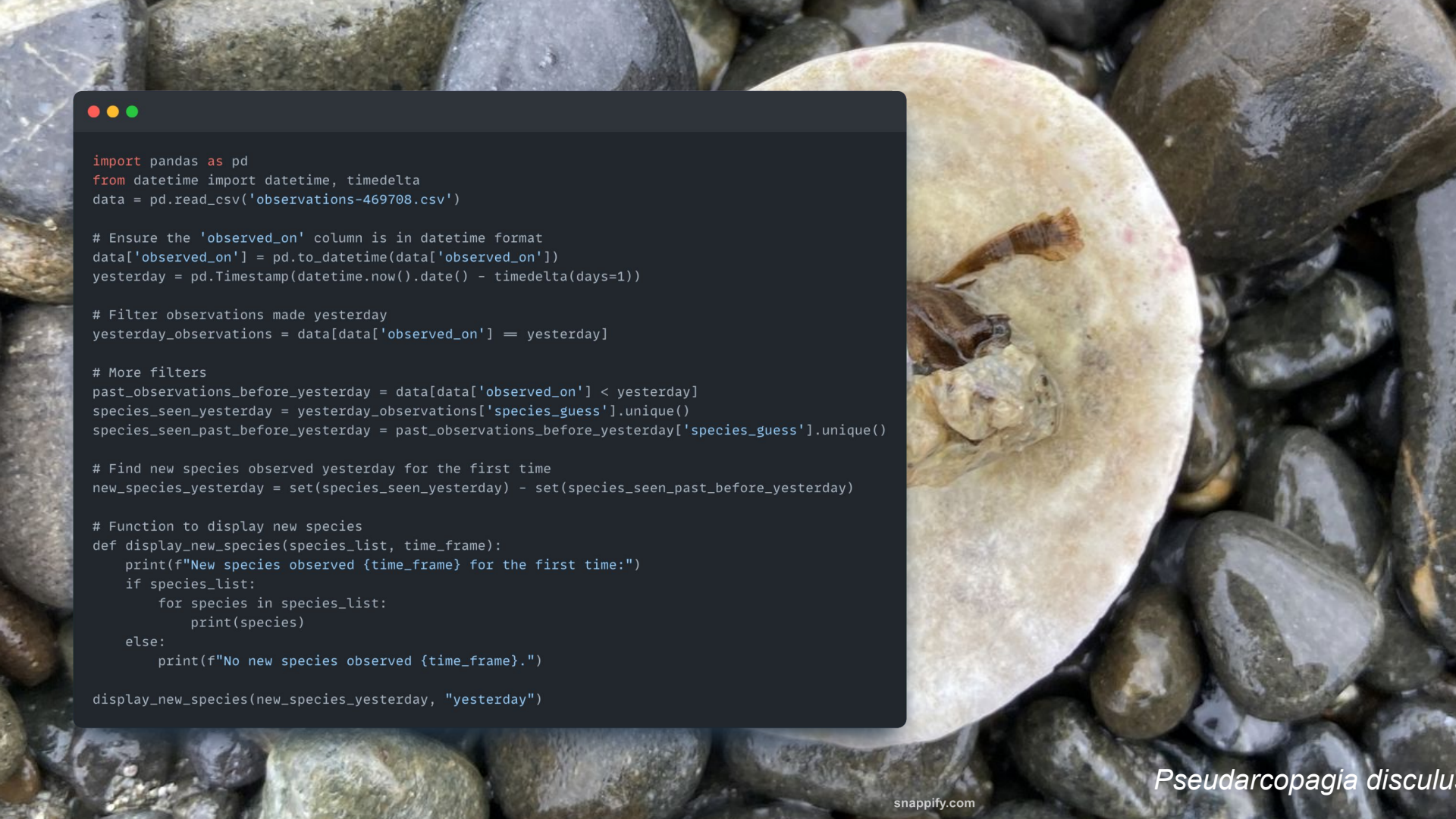
Hākēkeke

Give me new RG species

- Can't find this on iNat
- Useful to see what is new vs. forgotten



Pseudarcopagia disculu



```python
import pandas as pd
from datetime import datetime, timedelta
data = pd.read_csv('observations-469708.csv')

# Ensure the 'observed_on' column is in datetime format
data['observed_on'] = pd.to_datetime(data['observed_on'])
yesterday = pd.Timestamp(datetime.now().date() - timedelta(days=1))

# Filter observations made yesterday
yesterday_observations = data[data['observed_on'] == yesterday]

# More filters
past_observations_before_yesterday = data[data['observed_on'] < yesterday]
species_seen_yesterday = yesterday_observations['species_guess'].unique()
species_seen_past_before_yesterday = past_observations_before_yesterday['species_guess'].unique()

# Find new species observed yesterday for the first time
new_species_yesterday = set(species_seen_yesterday) - set(species_seen_past_before_yesterday)

# Function to display new species
def display_new_species(species_list, time_frame):
    print(f"New species observed {time_frame} for the first time:")
    if species_list:
        for species in species_list:
            print(species)
    else:
        print(f"No new species observed {time_frame}.")

display_new_species(new_species_yesterday, "yesterday")
```


snappify.com

Give me new RG species



Known also as *whare atua* (‘house of the spirit’), *kopi* (‘shut’), or *Hineraukatauri* (the goddess of Māori flute music).

https://ento.org.nz/blog/2022/07/27/common-bag-moth/

- This is how you learn.

Give me new RG species



New Zealand Common Bag Moth

(*Liothula omnivora*)

Research Grade 2 1d



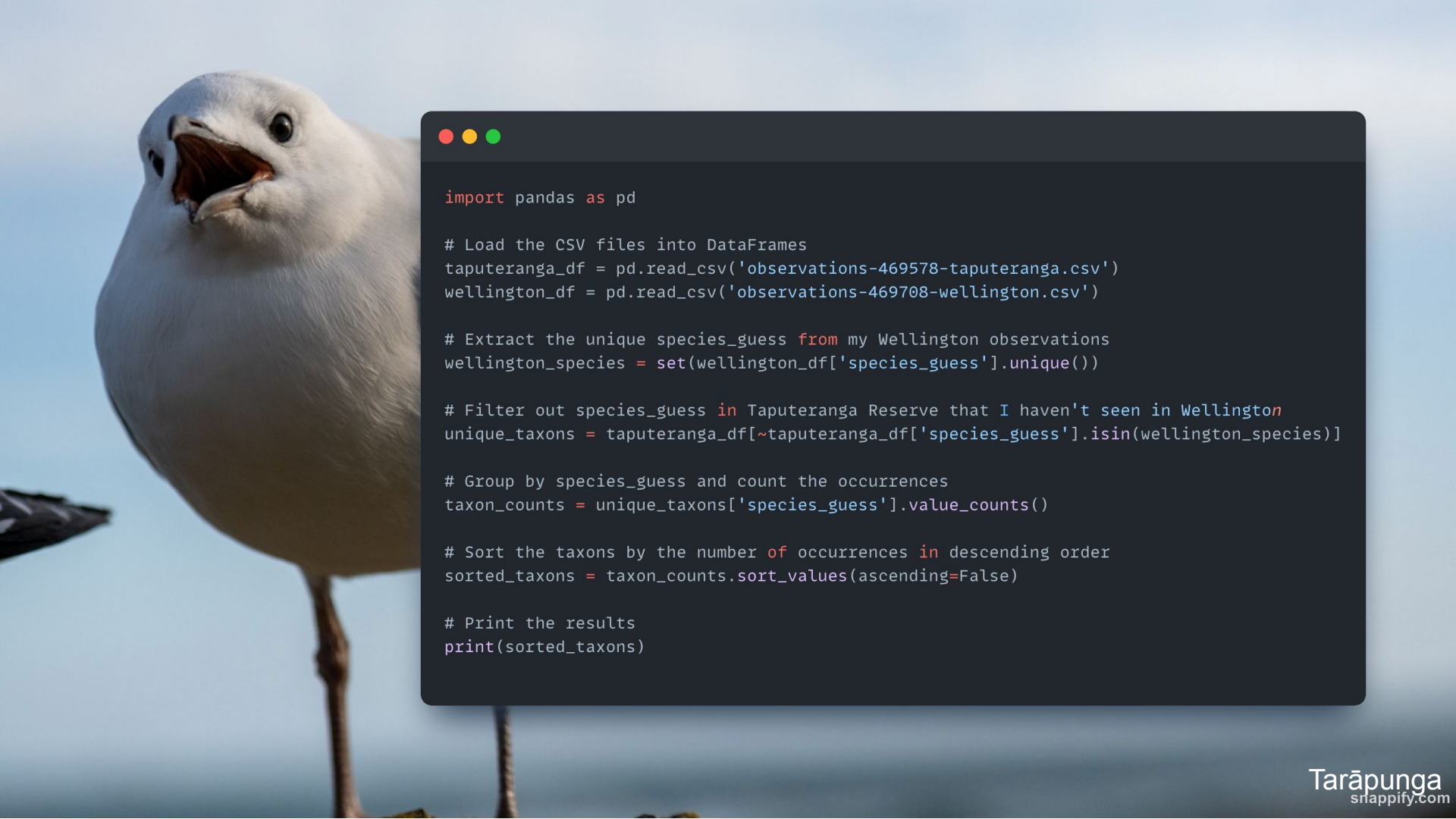
Trailing Treasureflower

(*Gazania rigens*)

Research Grade 2 1d

Introduced.


Find things you haven't seen
- Possible on eBird
- Possible on iNat (sorta)
- Filter other obs in an area
Tarāpunga



```python
import pandas as pd

# Load the CSV files into DataFrames
taputeranga_df = pd.read_csv('observations-469578-taputeranga.csv')
wellington_df = pd.read_csv('observations-469708-wellington.csv')

# Extract the unique species_guess from my Wellington observations
wellington_species = set(wellington_df['species_guess'].unique())

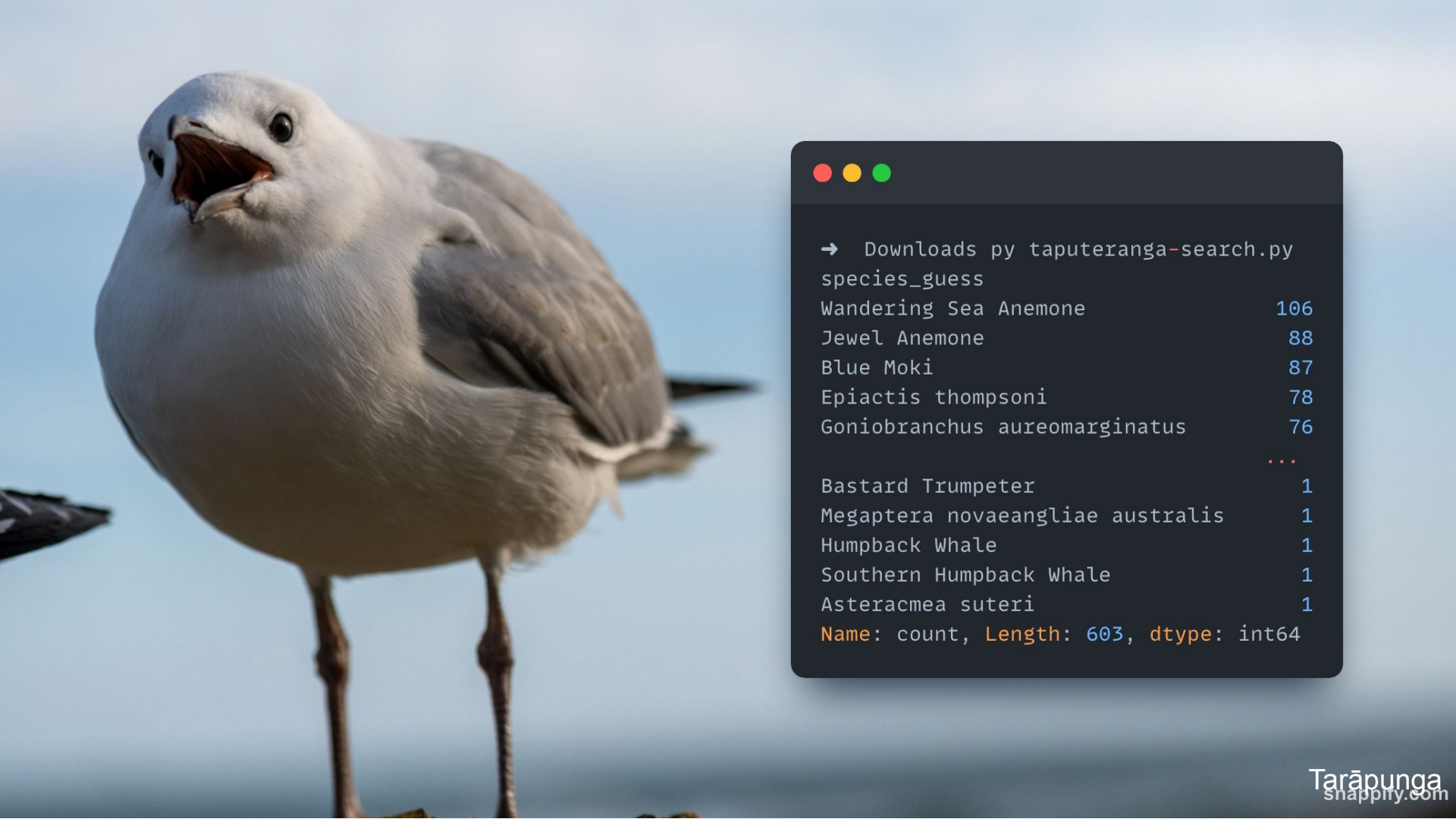
# Filter out species_guess in Taputeranga Reserve that I haven't seen in Wellington
unique_taxons = taputeranga_df[~taputeranga_df['species_guess'].isin(wellington_species)]

# Group by species_guess and count the occurrences
taxon_counts = unique_taxons['species_guess'].value_counts()

# Sort the taxons by the number of occurrences in descending order
sorted_taxons = taxon_counts.sort_values(ascending=False)

# Print the results
print(sorted_taxons)
```

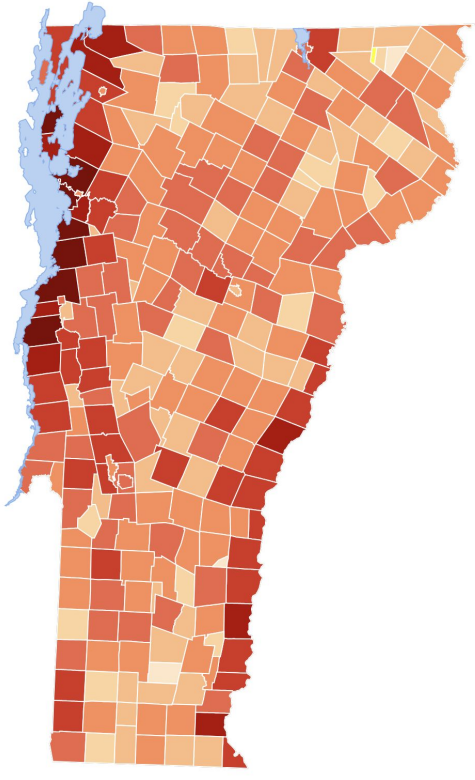
Tarāpunga
snappify.com



```
➜  Downloads py taputeranga-search.py species_guess
Wandering Sea Anemone                106
Jewel Anemone                         88
Blue Moki                             87
Epiactis thompsoni                    78
Goniobranchus aureomarginatus         76
                                     ...
Bastard Trumpeter                      1
Megaptera novaeangliae australis       1
Humpback Whale                         1
Southern Humpback Whale                1
Asteracmea suteri                      1
Name: count, Length: 603, dtype: int64
```

Tarāpunga
snappify.com

JavaScript and Bash?



Warner's Grant (13)

Seen

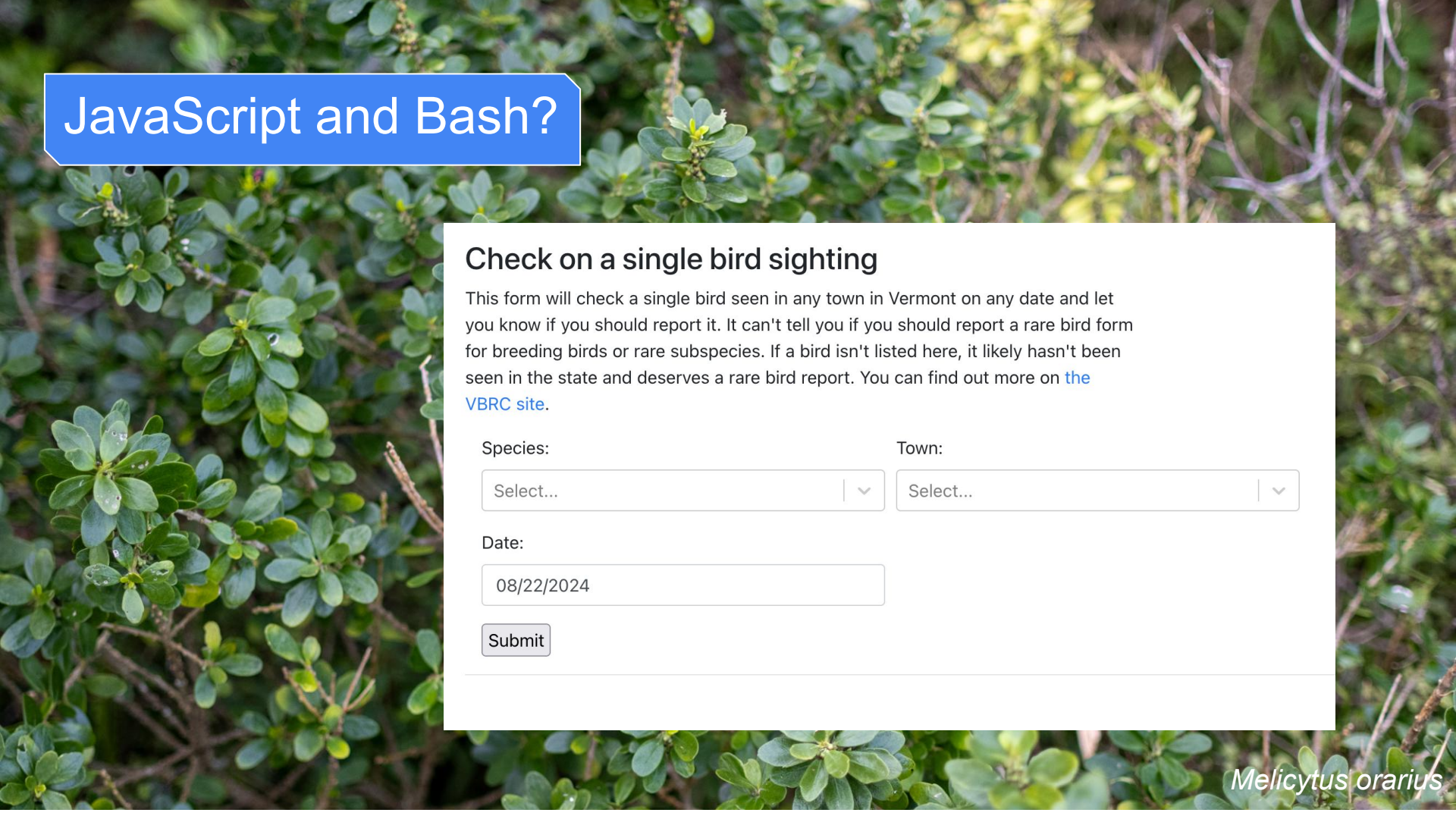
- Blue Jay
- Black-throated Blue Warbler
- Black-throated Green Warbler
- Eastern Wood-Pewee
- Least Flycatcher
- Ovenbird
- Red-eyed Vireo
- Tufted Titmouse
- American Goldfinch
- Canada Goose
- Common Raven
- Downy Woodpecker
- White-breasted Nuthatch

No records:

- Black-bellied Whistling-Duck
- Fulvous Whistling-Duck
- Snow Goose
- Ross's Goose
- Greater White-fronted Goose
- Pink-footed Goose
- Brant
- Barnacle Goose
- Cackling Goose
- Mute Swan
- Trumpeter Swan
- Tundra Swan
- Wood Duck
- Garganey
- Blue-winged Teal

Melicytus orarius

JavaScript and Bash?



Melicytus orarius

JavaScript and Bash?

GET Nearby hotspots

`https://api.ebird.org/v2/ref/hotspot/geo?lat={{lat}}&lng={{lng}}`

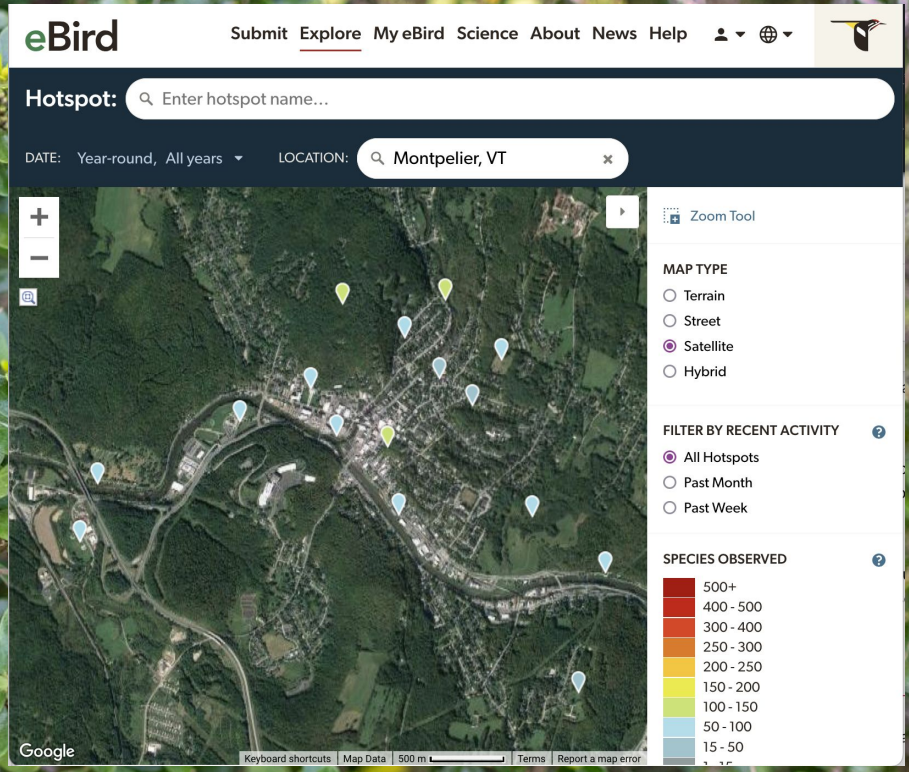
Get the list of hotspots, within a radius of up to 50 kilometers, from a given set of coordinates.

Query parameters

Name	Values	Default	Description
back	1-30	(none)	Only fetch hotspots which have been visited up to 'back' days ago.
dist	0 - 500	25	The search radius from the given position, in kilometers.
fmt	csv, json	csv	Fetch the records in CSV or JSON format.
lat	-90 - 90		**Required.** Latitude to 2 decimal places.
lng	-180 - 180		**Required.** Longitude to 2 decimal places.

AUTHORIZATION API Key

This request is using API Key from collection eBird API 2.0



Melicytus orarius

JavaScript and Bash?

```python
#!/usr/bin/env python3

import os
import sys
import email
import getpass
import smtplib, ssl

from email.message import EmailMessage

content_file = '/Users/richard/src/scripts/logs/hotspots.log'

PORT = 465

with open(content_file, 'r', newline='') as f:
    content = f.read()

content = "Hey there,\n" + content + "\nI have the honour to be your obedient servant,\nRichard Littauer"

def send_email(content, sender='richard@burntfen.com', receiver='richard+birding@burntfen.com'):
    context = ssl.create_default_context()
    password = os.environ.get('GMAIL_PASS')
    smtp_server = "smtp.gmail.com"
    message = EmailMessage()
    message.set_content(content)
    message["Subject"] = "Hotspot Coverage in Montpelier Today"
    message["From"] = sender
    message["To"] = ', '.join(receiver)

    with smtplib.SMTP_SSL(smtp_server, PORT, context=context) as server:
        server.login(sender, password)
        server.send_message(message)

if __name__ == "__main__":
    print(content)
    send_email(content)
    print("Done! Email sent!")
```

Melicytus orarius

snappify.com


PhD work?
Kākā

Why is this a Python talk



?

Thanks



http://richard.social
inaturalist.org
eBird.org

Weka